Beyond the edge of stability: periodic windows, quasi-periodicity and phase locking in the learning-rate plane of gradient descent

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Abstract

Large-step gradient descent can sustain oscillatory training after fixed-point convergence becomes unstable. We study how these dynamics are organized in learning-rate planes of small neural networks, combining full-state orbit refinement, stability calculations and exploratory bifurcation analysis.

A prospectively designed map study, independently cross-checked in a separate numerical implementation, identifies 61 attracting periodic windows in two tanh networks with separate layer learning rates. Their surrounding cells pass finite-horizon chaos diagnostics. Among cells labeled convergent by a published loss-reduction score, 87.1% to 99.97% are not classified as fixed points after 20,000 updates. The score therefore combines several dynamical regimes.

Exploratory calculations identify a local organizing mechanism at selected period-two orbits: a complex pair of multipliers crosses the unit circle, with a rotation angle that predicts the ensuing oscillation frequency. The first Lyapunov coefficient predicts soft versus hard onset in 8 of 8 tested cases. Related crossings occur in a network trained on digit images and in low-dimensional factorization models. An explicit scalar control isolates how unequal rates permit complex two-step dynamics. Near one predicted resonance, a locked region narrows with the expected asymptotic scaling, but its connection to a selected distant window remains unresolved and the prespecified ancestry test did not pass.

Rotation features provide no robust additional prediction of loss or disagreement magnitude in the tested pilots; adapted analyses retain locking effects on perturbation response and learning-rate-decay timing. Loss-based estimation of the local complex multiplier is a candidate diagnostic whose prospective utility remains untested. Separately, ball-arithmetic certificates prove local existence, primitive period and attraction for 2 selected period-10 orbits in a 13-parameter network and a period-2 orbit in a 72-parameter network, all for the exact-real maps. A separate certificate establishes a local supercritical Neimark–Sacker bifurcation in one scalar unequal-rate model. The results do not establish a general route to chaos, a language-model benefit, or computer-assisted proofs of the reported window census.

Study status. Results labeled PROSPECTIVELY DESIGNED, INDEPENDENTLY CROSS-CHECKED come from the map study of Section 4, whose design, thresholds and analysis were pushed before outcomes and checked by an independent numerical implementation. This describes design timing and cross-checking; it does not turn the subsequent exploratory mechanism analyses into confirmatory tests on fresh data. Results labeled EXPLORATORY (Section 6) were found while exploring; some used acceptance rules frozen before the run, and we say so where they did. Provenance for every figure and number is described in Section 10.

Component	Evidence rung	Scope and outstanding work
Periodic-window census	2: independently cross-checked	Floating-point orbit closure, primitive-period and attraction tests; finite-horizon chaos diagnostics in surrounding cells.
Local crossing and criticality	1: numerically validated	Tolerance and derivative checks in explored systems; confirmation on fresh losses remains pending.
Global resonance ancestry	0: exploratory/adapted	Near-root scaling is suggestive; the prespecified ancestry test did not pass and the distant-window connection is unresolved.
Training consequences	0–1: exploratory/adapted	Small-model slices and two controlled ramp points; no geometry-specific advantage over a generic rate reduction.
CIFAR and language-model transfer	0: exploratory development	CIFAR orbit correction remains unresolved after checkpointed solver failures; two text-transformer prediction pilots fail their incremental-utility gates. No accepted transformer orbit or training benefit.
Selected orbit certificates	3: computer-assisted proof	Two primitive period-10 orbits in 13 dimensions and one primitive period-2 orbit in 72 dimensions, all attracting for the exact-real maps; surrounding chaos and the census remain numerical.
Scalar bifurcation certificate	3: computer-assisted proof	One local supercritical NS of an exact-real primitive period-2 orbit; no finite-detuning curve enclosure or global ancestry.

Table 1: Evidence and scope. Design timing and mathematical rigor are separate: a prospective numerical study can be independently cross-checked without enclosing roundoff. No result here is formally verified in a proof assistant. Source receipts are indexed in Section 10.

1 Introduction

Full-batch gradient descent can continue reducing neural-network training loss beyond the local quadratic stability threshold. In the edge-of-stability regime observed by Cohen et al. [6], sharpness stays near $2/\eta$ while short-term loss fluctuations coexist with longer-term decrease. Related analyses address self-stabilization under local regularity and spectral assumptions [10], progressive sharpening in second-order regression models [1], and initial loss growth followed by convergence in the catapult regime [24]. These descriptions do not define a universal sequence of training phases. Simplified models show period doubling and chaos as the step size grows [5, 15, 20]. Numerical studies report fractal trainability or loss-convergence boundaries in selected learning-rate planes [26, 32, 35]. Related factorization results concern convergence regions in initialization space [25]. Such success criteria do not by themselves distinguish full-state dynamical regimes.

We ask what the dynamics inside such maps are. We treat full-batch gradient descent with separate learning rates for the hidden and output layers as a two-parameter family of maps on parameter space. For every cell of a learning-rate grid we screen the parameter state at the end of a long run, refine selected periodic orbits with Newton's method using an analytic Jacobian evaluated in floating-point arithmetic, and follow orbits and their stability boundaries through the plane.

Thesis and status. We organize the results around one thesis: beyond the edge of stability, the period-2 oscillation gives way to quasi-periodic training on an invariant circle, rational rotation numbers lock into periodic windows, and the windows of the learning-rate plane are largely these locked orbits. Each result below says exactly what passed. The cross-checked windows and the comparison with the binary score are confirmed. Exploratory analyses support the local part of the thesis: complex-pair crossings occur in independently initialized slices, the sampled crossings are mostly supercritical with dynamics that match the normal form, quasi-periodic training starts with a rotation equal to the crossing angle, and the same crossing appears in a real-data network and in a linear network. The global part is not established: a frozen test did not confirm that a rule-selected window descends from its resonance, and no prespecified path into sustained chaos has been tested. We call the crossings Neimark–Sacker candidates except where the nondegeneracy coefficient and dynamical checks were computed.

Prior work. Periodic windows, period doubling, positive Lyapunov exponents and broadband power spectra in learning dynamics, with the learning rate as bifurcation parameter, date back at least to van der Maas et al. [36], who studied a three-unit autoassociator. Their delay-embedded chaotic attractors are thin folded curves, effectively one-dimensional chaos of the period-doubling kind, unlike the invariant circles of the two-step map studied here. Period-colored maps over optimizer hyperparameters have been drawn from a two-dimensional projection of the error, without full-state verification [34]. Hofmann [15] studies finite-step gradient descent as a dynamical system in solvable models: the edge of stability as the first (flip) bifurcation of the training map, the period-doubling route with windows, and, in a two-mode depth-two linear network, a "rotational edge" beyond which the unstable multipliers of the aligned 2-cycle are complex and the system saturates in a persistently misaligned oscillatory state. The edge is Hofmann's result; we show, in that model and in nonlinear networks, numerical evidence for post-edge closed curves whose rotation approaches the crossing angle, and for locking; the linear model requires explicit care with its continuous symmetry. Empirical riddling-like sensitivity and learning-rate-dependent reproducibility under nearby-initialization retraining are reported by Ly and Gong [27]. Invariant curves and phase-locking Arnold tongues appear in a multiplicative-weights learning rule for population games [28], and chaos in learning a two-person game goes back to Sato et al. [31]; both concern games

rather than descent on one loss. For heavy-ball momentum, Goujaud et al. [14] construct cycles on smooth strongly convex two-dimensional objectives that advance by $1/K$ of a turn per update.

This paper's contribution.

- Confirmed: 61 numerically validated attracting windows embedded in chaos in the learning-rate planes of two small networks, repeated by an independent audit, and a cell-by-cell comparison showing that the published binary score conflates fixed points, periodic orbits, chaos and drift.
- Exploratory, local mechanism: a merged locus of slice-wise period-two complex-pair crossings; pointwise first Lyapunov coefficient, candidate generalized Neimark–Sacker points and dynamical checks; rotation-number maps in which quasi-periodic training begins at the crossing angle.
- Exploratory, generality: the same crossing with a quantitatively predicted rotation in a real-data network with one learning rate; soft onset and locking in symmetry-reduced observables beyond the rotational edge of a linear network; distinct continuation outcomes from two bootstrap procedures and a numerically consistent local period-2 segment in a 72-parameter network.
- Computer-assisted proofs from exploratory selection: two primitive attracting period-10 orbits in 13 dimensions, one primitive attracting period-2 orbit in 72 dimensions, and one local supercritical NS of a scalar period-2 orbit, independently reviewed with ball arithmetic.
- Reported negatives: a frozen ancestry test that did not pass; no locking resolved in the real-data network; no robust incremental value of rotation information for reproducibility or loss.
- Code, receipts and a verification script that re-derive every figure, table and number.

Not claimed.

- No route to chaos: no prespecified connected path through sustained chaos has been tested.
- No proof that the windows are Arnold tongues: the Birkhoff test is a diagnostic of projected orbits, and the frozen ancestry test did not pass.
- No claim about stochastic, minibatch or large-scale training, or other architectures and data beyond those shown, and no training benefit.

2 Gradient descent as a family of maps

Let $L(\theta)$ be the full-batch training loss of a network with parameters $\theta \in \mathbb{R}^n$, split into hidden-layer and output-layer blocks. Gradient descent with per-layer learning rates is the map

$$\theta_{t+1} = F(\theta_t) = \theta_t - P\,\nabla L(\theta_t), \qquad P = \mathrm{diag}(\eta_h \mathbf{1}_{\mathrm{hidden}}, \eta_o \mathbf{1}_{\mathrm{output}}), \tag{1}$$

a family of maps indexed by the two rates (η_h, η_o). Its Jacobian is $DF(\theta) = I - PH(\theta)$ with H the Hessian of L. An orbit of period p is stable when all eigenvalues (multipliers) of $DF(\theta_{p-1}) \cdots DF(\theta_0)$ lie inside the unit circle. Generically it loses stability when a multiplier crosses the circle: at $+1$ (a fold candidate, subject to nondegeneracy and symmetry checks), at -1 (a flip candidate) or as a

complex pair $e^{\pm 2\pi i\omega}$ (a Neimark–Sacker candidate). For a simple transverse complex crossing with all other multipliers strictly inside the unit circle, no strong resonance $e^{2\pi ik\omega} = 1$ for $k = 1, 2, 3, 4$, and nonzero first Lyapunov coefficient d, a Neimark–Sacker bifurcation creates an invariant closed curve: attracting with amplitude growing like the square root of the distance beyond the crossing when $d < 0$ (supercritical), repelling and present before the crossing when $d > 0$ (subcritical) [23].

Why the candidates start at period 2. $DF(\theta)$ is similar to the symmetric matrix $I - P^{1/2}HP^{1/2}$, so at a fixed point all multipliers are real and local stability can be lost only through a real unit-circle crossing at $+1$ or -1. The $+1$ case need not be a fold; symmetry and other degeneracies can produce different bifurcations. At a nondegenerate minimum under a common positive scaling of P, the first instability is a -1 crossing. The edge-of-stability period-2 branch is associated with that flip. For a period-2 orbit $\{\theta_a, \theta_b\}$, the two-step Jacobian is similar to S_bS_a with $S_j = I - P^{1/2}H(\theta_j)P^{1/2}$. A product of two symmetric matrices has real eigenvalues when one factor is positive definite (also when one is negative definite). Along post-edge cycles the factors can instead be indefinite and fail to commute, allowing complex multipliers. Indefiniteness alone is not a sufficient condition. This is an elementary observation; complex unstable multipliers of an edge-of-stability 2-cycle were derived for a two-mode linear network by Hofmann [15].

Circle maps and phase locking. On an invariant circle the dynamics has a rotation number ρ, the mean fraction of a turn per iterate. Rational rotation m/q can produce a phase-locked periodic orbit. Near a nondegenerate weak resonance ($q \geq 5$), an appropriate resonant normal form predicts a tongue whose width in angular detuning scales with radial detuning to the power $(q-2)/2$, when the leading resonant coefficient is nonzero. This local scaling does not fix the width along arbitrary distant parameter slices. Analytic circle maps sufficiently close to rotation admit analytic conjugacy to a rigid rotation under arithmetic conditions on their irrational rotation number [2]. Classical circle-map families exhibit devil's staircases [16, 17]; these are hypotheses to test in a learning-rate plane, not automatic consequences of a complex-pair crossing. Breakdown of an invariant circle is one possible route to chaos [3, 9, 29]. Locally near the period-2 bifurcation, F exchanges two disjoint components and a period-q locked orbit of F^2 has primitive period $2q$ for F. This relation follows from the local setting and is not independent evidence of resonance ancestry.

Optimizer state and momentum. Our map study concerns plain gradient descent. For standard heavy ball, $x_{t+1} = x_t - \eta\nabla L(x_t) + \beta(x_t - x_{t-1})$, each Hessian eigenvalue h at a stationary point gives

$$\mu^2 - (1 + \beta - \eta h)\mu + \beta = 0. \tag{2}$$

At fixed $0 < \beta < 1$, a complex pair has modulus $\sqrt{\beta} < 1$ and cannot cross the unit circle as η varies. A nondegenerate minimum first loses local stability through a flip at $\eta h_{\max} = 2(1 + \beta)$. A nonsymmetric augmented Jacobian therefore does not, by itself, imply a fixed-point Neimark–Sacker mechanism. Periodic branches and adaptive optimizers require their own full-state analysis, including moment variables and decay. Bias correction, schedules and minibatches introduce time dependence or randomness; an instantaneous Jacobian along such a trajectory is not automatically a Floquet matrix.

Shrimps. In two-parameter families of dissipative maps, periodic windows inside chaos often take a characteristic "shrimp" shape, with a primary periodic body bounded by fold and flip curves and adjoining period-doubling regions [12, 13]. We test for this anatomy explicitly rather than assume it.

3 Methods

Network	Parameters	Training points	Architecture	Weight decay	Seeds	Maps
S1	72	72 standard-normal	8 inputs, 8 tanh units, no biases, mean-field	0	28001, 28002	2 coarse
S1R	72	72 standard-normal	8 inputs, 8 tanh units, no biases, mean-field	10^{-3}	28001, 28002	2 coarse, 2 zoom
S2	13	24 from a tanh teacher	2 inputs, 3 tanh units, biases	10^{-3}	28011, 28012	2 coarse, 2 zoom

Table 2: Design of the map study. Every map has 256×256 logarithmically spaced pairs of hidden- and output-layer rates; coarse maps span 3 decades in each rate and zoom maps 0.36 decades. Every cell runs 20,000 full-batch updates in float64 from its seed's initial parameters. S1 is the published configuration of the fractal-trainability study and is reported descriptively.

Networks and data. Table 2 lists the design of the map study. S1 reproduces the published mean-field configuration of the fractal-trainability study [32]; S1R adds weight decay; S2 is a 13-parameter tanh regressor fitted to points from a random tanh teacher. We refer to a network and seed together, for example S2-28011. Training is deterministic full-batch gradient descent in float64.

Three different quantities. The published binary score measures loss reduction: a run counts as convergent when its loss, normalized by the initial loss and averaged over the last 20 of the first 1,000 updates, is below one [32]. Our screening labels classify the parameter state at the end of a finite run. Orbit refinement tests individual periodic orbits against numerical tolerances. The three need not agree, and we compare them cell by cell.

Screening. Every cell runs 20,000 updates from the seed's initial parameters. The last 384 updates are screened for the smallest full-state recurrence period up to 64: a period p requires the state to recur with componentwise tolerance $10^{-8} + 10^{-6}\,|\theta|$ over the whole tail, which spans at least 6 repeats. The largest finite-time Lyapunov exponent (FTLE) is averaged over the last 4,000 updates after 1,000 updates of tangent alignment. Labels are fixed point, periodic, chaotic (FTLE above 0.005 per update), drift (parameter norm still growing by more than 1% over the FTLE window), unresolved, or divergent (non-finite or larger than 10^8).

Numerically validated. An orbit of period p is numerically validated when damped Newton iteration on $F^p(\theta) - \theta = 0$, using the exact chain-rule Jacobian of F^p, reaches an undamped relative correction below 10^{-10} within 60 iterations; the converged point closes to within $10^{-9} + 10^{-8}\,|\theta|$ in every component; it lies within 10^{-3} (relative) of the screened candidate; every proper-divisor iterate is separated from it by at least 10^{-6} (relative), so the period is primitive; and it is attracting when the spectral radius of the monodromy matrix is below $1 - 10^{-7}$. These checks use floating-point arithmetic and stated tolerances; they do not enclose rounding error. A reviewed correction to complex left-eigenvector pairing affects a separate sensitivity diagnostic. The audit found no affected saved orbit decision or criticality coefficient; original records and the precise exception-handling limitation are preserved in `docs/complex-left-condition-impact.md`.

Embedded windows and audit. A connected component of equal-period cells qualifies as an embedded window when its period is at least 3, it has at least 4 cells, no cell of its exterior ring is a fixed point or has a proper-divisor period, at least 50% of the ring is rerun-confirmed chaos (reruns of 30,000 updates with a 10,000-update FTLE and a different tangent vector, up to 48 ring cells per component), and its representative orbit passes the numerical orbit and attraction checks. The audit was a separate program that imports neither the batched engine, the orbit-refinement implementation nor the selection helpers: it rewrote the one-cell map, the recurrence screen, the zoom rule and the component and ring rules, rechecked every numerical orbit record with finite-difference Jacobians (all 154 of 154 pass closure, primitivity, stability and seed checks; spectral radii agree within 1.7×10^{-7}), re-simulated the ring cells of every qualifying component, recomputed the gate on its own evidence, and found 0 mismatches among the 276 receipt files.

Continuation, boundary type and criticality. From the center of a window, its numerically validated orbit is continued along 64 rays in the rate plane until it loses stability, located by bisection, and the boundary is typed by the crossing multiplier: fold candidate ($+1$), flip candidate (-1) or complex-pair candidate. These multiplier labels alone do not establish the corresponding nondegeneracy conditions. The same procedure traces where the period-2 orbit loses stability along slices of fixed output rate. Each slice is initialized separately; sorting and deduplicating their crossings gives a merged locus, without establishing full-state branch identity across output rates. At each selected slice-wise crossing we compute the first Lyapunov coefficient d of the Neimark–Sacker normal form with Kuznetsov's formula for n-dimensional maps, using the analytic dense two-step Jacobian evaluated in float64 and Richardson-extrapolated differences of analytic Jacobian-vector products for the higher derivatives; the implementation reproduces the explicit planar formula on random planar maps.

Rotation numbers and the Birkhoff test. For a non-periodic cell we take the two-step samples $z_k = \theta_{2k}$, project them on their first two principal axes, and define the rotation number as the mean unwrapped angle increment in turns per two steps. Rotation-number maps run this on every cell of a grid (float64 on a GPU, 20,000 screening updates followed by 6,000 two-step samples). For a numerically validated orbit of period p we use F^2 for even p ($q = p/2$ points) and F for odd p ($q = p$ points), sort them by angle in their principal plane, and say the orbit passes when every step advances the angular rank by the same shift m with $\gcd(m, q) = 1$, the ordering of a Birkhoff orbit of rotation m/q. The number of points q is fixed by the period and the chosen iterate; the relation $p = 2q$ applies only to the even-period case; what the test measures is the ordering and the shift. Orbits with 2 or 3 points pass by construction. The null is the pass probability of the same test for uniformly random angles; for a group of orbits we report the expected number of chance passes. The refined test also orders points along the shortest closed tour through them and tests the points of $F^{2^{k+1}}$ ($k = 1, 2, 3$), which can detect orbits born by period doubling inside a locked window; it counts only tests on at least 5 points, and its null for each orbit is the sum of the per-test pass probabilities (a union bound), so the summed null is an upper bound on the expected number of chance passes.

Resonance ancestry. For a window selected by a fixed rule (the 1/5 window of S2-28011), we locate the points of the crossing curve whose angle is 1/5, check the onset there, and follow the locked period-10 interval on six cross-sections at distances $0.0128/4^k$ beyond the curve, finding both of its fold edges on each. Acceptance rules were frozen before the network run: controls must pass first; the interval must be fold-bounded on every level with shrinking width and amplitude; and a

linear fit of its center over all levels must extrapolate to within 10^{-4} decades of the predicted point. A second stage tries to connect the tracked interval to the window.

4 Periodic windows inside chaotic training

Status: PROSPECTIVELY DESIGNED, INDEPENDENTLY CROSS-CHECKED.

The learning-rate planes of S1R and S2 contain 61 embedded windows that qualify on both the producer's and the auditor's evidence (62 and 63 respectively; the disagreements all sit at the 50% ring threshold). All 154 attempted orbit refinements passed the specified numerical checks. Confirmed periods range from 4 to 58 and include the odd periods 5, 11, 13, 17 and 29. The exterior rings were checked with 2,801 reruns, and 157 of 160 randomly chosen chaotic cells (16 per map) stayed chaotic on a rerun with a different tangent vector.

Figure 1 compares these dynamics with the published binary score. In the six coarse maps, 87.1% to 99.97% of the cells the score labels convergent were not classified as fixed points after 20,000 updates; the score's success means the loss fell, not that the parameters settled. Task success and non-convergent learning dynamics can coexist: van der Maas et al. [36] reported that their network performed its task across the whole learning-rate range they studied, including where learning was chaotic. The published configuration without weight decay (S1) has 0 periodic cells in both seeds: its runs diverge, stay chaotic, or drift toward large weights while the loss slowly falls. Adding weight decay makes the windows appear.

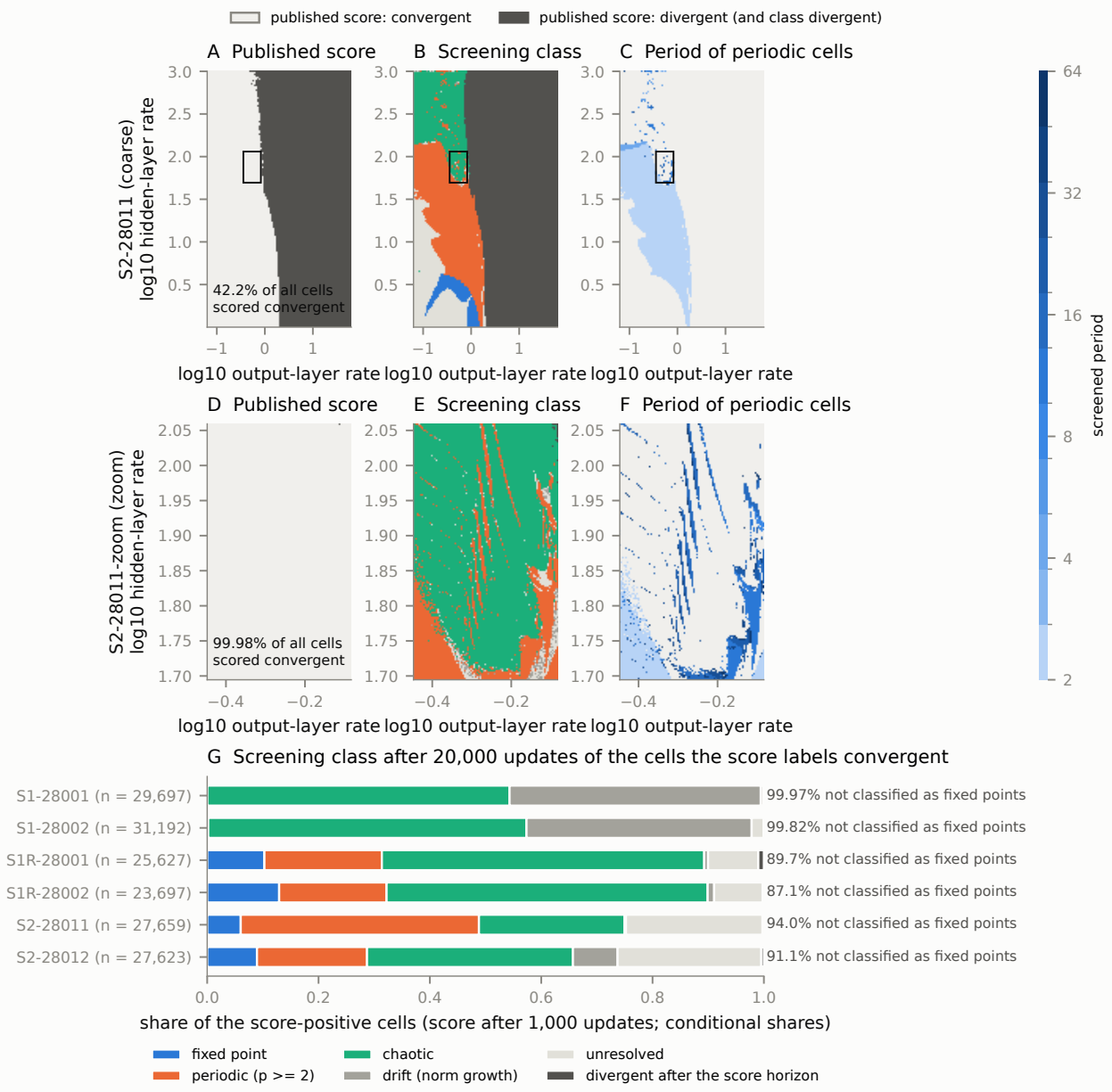


Figure 1: The binary score conflates dynamical regimes: the cells it labels convergent carry fixed-point, periodic, chaotic and drift labels after 20,000 updates. Panels A and D show the published score after 1,000 updates; the printed share is the fraction of all cells in that map scored convergent. Panels B, C, E and F show the screening class and, for periodic cells, the period of the same cells after 20,000 updates (S2-28011, coarse and zoomed; the rectangle marks the zoom). Panel G is conditional: among the cells the score labels convergent, the share in each screening class, for all six coarse maps. Screening labels are finite-time labels; only refined orbits receive numerical closure, primitivity and attraction checks. *Generated by* `paper/gd-route/figures/f1_binary_vs_period.py`; *receipt* `figures/f1_binary_vs_period.provenance.json`.

5 Computer-assisted proofs of selected orbits and a local bifurcation

Status: exploratory target selection, evidence rung 3 for the statements below.

5.1 Attracting periodic orbits

The floating-point census motivates a separate proof task. For 2 selected S2 settings, Arb ball arithmetic encloses a unique zero of $H(x) = F^{10}(x) - x$ in a specified initial box, excludes all proper divisors of 10, and proves attraction. One target, S2-28011 component 140, belongs to the historically qualifying window set. The other, S2-28011-zoom component 936, was the initial feasibility choice; it does not qualify for that historical window claim. Its valid orbit proof is retained with that distinction.

The theorem concerns real arithmetic with the supplied dataset, rates, ridge and loss scale interpreted as exact dyadic constants. It does not assert a periodic orbit of the binary64 implementation. An analytic ball Jacobian encloses the derivative throughout the initial box and its iterated images. With an exact dyadic approximate inverse R, strict Krawczyk inclusion and $\|I - RH'(X)\|_\infty < 1$ establish contraction of $x \mapsto x - RH(x)$ on X and the nonsingularity of R. Disjoint coordinate enclosures of $F^d(X)$ and X exclude each proper divisor d. An interval monodromy M and repeated squaring give

$$\|M_{\text{window}}^{64}\|_\infty < 0.056 \quad \text{and} \quad \|M_{\text{feas}}^{64}\|_\infty < 0.108, \tag{3}$$

respectively, implying spectral radius below one at each enclosed orbit. The initial sequential-power attempt failed to enclose attraction because of interval wrapping; that inconclusive attempt and the disclosed repeated-squaring amendment remain available.

An independent review reproduced the certificates, checked historical input bindings and stored witnesses, compared the analytic derivatives with separate second-order automatic differentiation of the objective, and repeated the proofs with higher precision on smaller nested boxes. The proof trusts the mathematical contraction argument, implementation and Arb/FLINT enclosure library; it is not formalized in a proof assistant. It establishes neither surrounding chaos nor window geometry, invariant circles or training utility. The authoritative artifacts and offline replay are documented in `certificates/README.md`.

The same unchanged checker also proves a primitive period-2 orbit in the full 72-dimensional S1R-28002 map, selected from the saved attracting segment described below. For its enclosed two-step monodromy,

$$\|M_{\text{S1R}}^{2048}\|_\infty < 0.556 < 1. \tag{4}$$

The independent outcome audit binds the original data, rates and selected phase to their exact dyadic interpretations, replays every proof enclosure, and passes a separately declared whole-box proximity test to that saved phase. Proximity does not assert that the saved floating-point state lies inside the proof box, nor does it prove branch identity or continuous parameter continuation. This audit shares the previously reviewed Arb checker; the separate differentiation and higher-precision nested-box checks above concern the two S2 proofs. The new certificate extends the demonstrated proof dimension, without establishing a general scaling law or a proof of the S1R bifurcation. See `certificates/s1r-period2-receipt.json` and `docs/reviews/s1r72-certificate-outcome.md`.

5.2 A supercritical NS in the scalar unequal-rate model

Consider $L(u,v) = \frac{1}{2}(uv-d)^2 + \frac{\lambda}{2}(u^2+v^2)$, with

$$d = \frac{5404319552844595}{2^{52}}, \qquad \lambda = \frac{5404319552844595}{2^{54}}, \qquad (\eta_u, \eta_v) = (\eta, 4\eta). \tag{5}$$

These are the exact dyadic interpretations of the pilot's nominal target 1.2 and ridge 0.3. This selected case is separate from the illustrative amplitude-scaling example in Section 6.8. Let $G_\eta = F_\eta^2$. A three-variable Krawczyk inclusion encloses a unique root of $(G_\eta(x) - x, \det DG_\eta(x) - 1) = 0$ in the recorded box, whose coordinate radius is $2^{-40} + 2^{-69}$. Disjoint enclosures of $F_\eta(x)$ and x prove primitive period two. The crossing rate obeys $\eta_* \in [0.54834687232917, 0.54834687233100]$.

At that root the complete planar monodromy has determinant exactly one and trace in $[-0.79587626, -0.79587625]$; the trace enclosure excludes $-2, -1, 0, 2$, hence resonances of orders one through four. Invertibility of $DG_\eta - I$ gives an implicit local orbit branch. Along it,

$$\begin{aligned} \left.\frac{d|\mu|}{d\eta}\right|_{\eta_*} &\in [3.3529863, 3.3529864] \subset (0, \infty), \\ \ell = \mathrm{Re}(\overline{\mu} c_1) &\in [-1.700203, -1.700201] \subset (-\infty, 0). \end{aligned} \tag{6}$$

The right eigenvector has Euclidean norm one and its left row is normalized to pair to one with it and to zero with its conjugate. Tensor/resolvent and planar scalar-coefficient formulas both enclose a negative ℓ. Together these bounds meet the generic supercritical NS theorem hypotheses [23] for this polynomial map.

For $\eta > \eta_*$ sufficiently close to the crossing, the theorem supplies a qualitative local attracting invariant-curve consequence for G_η; under F_η the two local components are exchanged. It supplies no specified finite detuning, irrational rotation, global ancestry, basin size or training benefit. The certificate concerns exact-real dynamics and trusts Arb/FLINT and the stated theorems. Independent review uses rational polynomial differentiation, higher-precision nested enclosures and a separate cubic normal-form calculation. The source-bound receipt and scope are in `certificates/scalar-ns-receipt.json` and `certificates/scalar-ns-design.md`. The review separately checked the original seed binding; the frozen replay alone does not authenticate arbitrarily rewritten provenance metadata. This limitation and the independent checks are recorded in `docs/reviews/scalar-ns-independent.md`.

Exact rescaling of the same case. Let d_0 denote the certified target above. For $d > 0$, $\lambda = qd$, fixed rate ratio r, and $S_k(x) = kx$ with $k = \sqrt{d/d_0}$, substitution in the cubic GD map gives

$$F_{d,qd;\eta,r} \circ S_k = S_k \circ F_{d_0,qd_0;k^2\eta,r}. \tag{7}$$

Thus the certificate at $q = 1/4$, $r = 4$ transfers to every $d > 0$ on that ray, with $\eta_*(d) = \eta_*(d_0)d_0/d$ and orbit coordinates scaled by k. At corresponding phases the monodromy is unchanged. The total modulus slope along the orbit branch scales by k^2, and the cubic coefficient under the same unit-eigenvector convention scales by k^{-2}: second and third state derivatives of F^2 scale by k^{-1} and k^{-2}, respectively. This is one dimensionless case in $(\lambda/d, \eta d, r)$, not an independent replication or robustness to varying q or r. The algebra and independent review are in `docs/scalar-similarity-corollary.md`.

6 Exploratory evidence for the thesis

Status: EXPLORATORY. The analyses in this section were chosen while exploring and used data that shaped the thesis, except where frozen acceptance rules are stated.

6.1 Quasi-periodic training begins where the period-2 orbit's complex pair crosses the unit circle

Within each fixed-output slice, we continued an independently initialized period-two orbit of S2-28011 toward larger hidden-layer rates. The merged locus contains 69 slice-wise complex-pair crossings; continuous full-state branch identity between slices remains unverified. The crossing angle runs from 0.1670 to 0.2987 turns per two steps. Two rotation-number maps cover the region beyond it (Figure 2). At 0.002 decades beyond the curve, the measured rotation number equals the crossing angle at the same output rate with median absolute difference 5.8×10^{-4} over 39 points in the main map (largest 4.7×10^{-3}) and 3.6×10^{-4} over 31 points in the right-wing map (largest 2.7×10^{-3}). Of the resonance points inside each map, 5 of 7 and 5 of 6 have locked cells of the same rotation nearby; the others lie where the maps show no locked region of that rotation. In an independent staircase computation on the slice through the period-18 window, the curve places the crossing at $\log_{10} \eta_h = 1.738$ with angle 0.2180, and the first quasi-periodic cell lies at 1.738 with rotation 0.2180.

6.2 The crossing is mostly supercritical, and its dynamics match the normal form

At 62 of the 69 sampled crossings the first Lyapunov coefficient is negative, so the crossing is supercritical under the generic hypotheses; tested onset trajectories are consistent with the attracting-circle prediction (Figure 3). Negative criticality alone does not exclude phase locking on that circle. At 7 points, at both ends of the sampled output range, it is positive. Adjacent output-ordered rows have opposite signs in three places, near $\log_{10} \eta_o = -0.417$, $+0.005$ and $+0.053$. These nominate locations to investigate for generalized Neimark–Sacker (Chenciner) points. Full-state association across each bracket and the additional nondegeneracy conditions remain unverified. In 8 of 8 dynamical checks the behaviour matches the sign: at supercritical points the circle amplitude follows the square-root prediction (median relative error 0.4%, largest 5.5% next to an adjacent-row sign transition), and at subcritical points the state jumps away beyond the corresponding slice crossing while escape thresholds before it grow like the square root of the offset, with hysteresis. In the 72-parameter S1R network, the coefficient is negative at 15 of 16 sampled crossings, consistent with supercriticality under the same generic hypotheses.

The fixed-start maps show quasi-periodic training beyond sampled supercritical crossings. Their divergent arc cannot be identified with local subcriticality alone: at 7 sampled crossings the fixed-start map is divergent even just below the crossing, where the period-2 orbit is stable, and 4 of those points are supercritical, with a small attracting circle reached when training starts near the 2-cycle. These comparisons show that basin accessibility from the shared initialization matters in interpreting the divergent arc.

6.3 States along one slice

Figure 4 shows what training looks like along that slice. Beyond the crossing, the two-step samples lie on a closed curve (rotation 0.2159, FTLE 0.0001); inside the period-18 window they are nine points (rotation 0.2222, FTLE -0.0072); between the 2/9 and 1/4 plateaus the projected state is irregular and has positive finite-time growth (FTLE 0.0504). These separate sampled trajectories do not establish a connected invariant-circle breakdown.

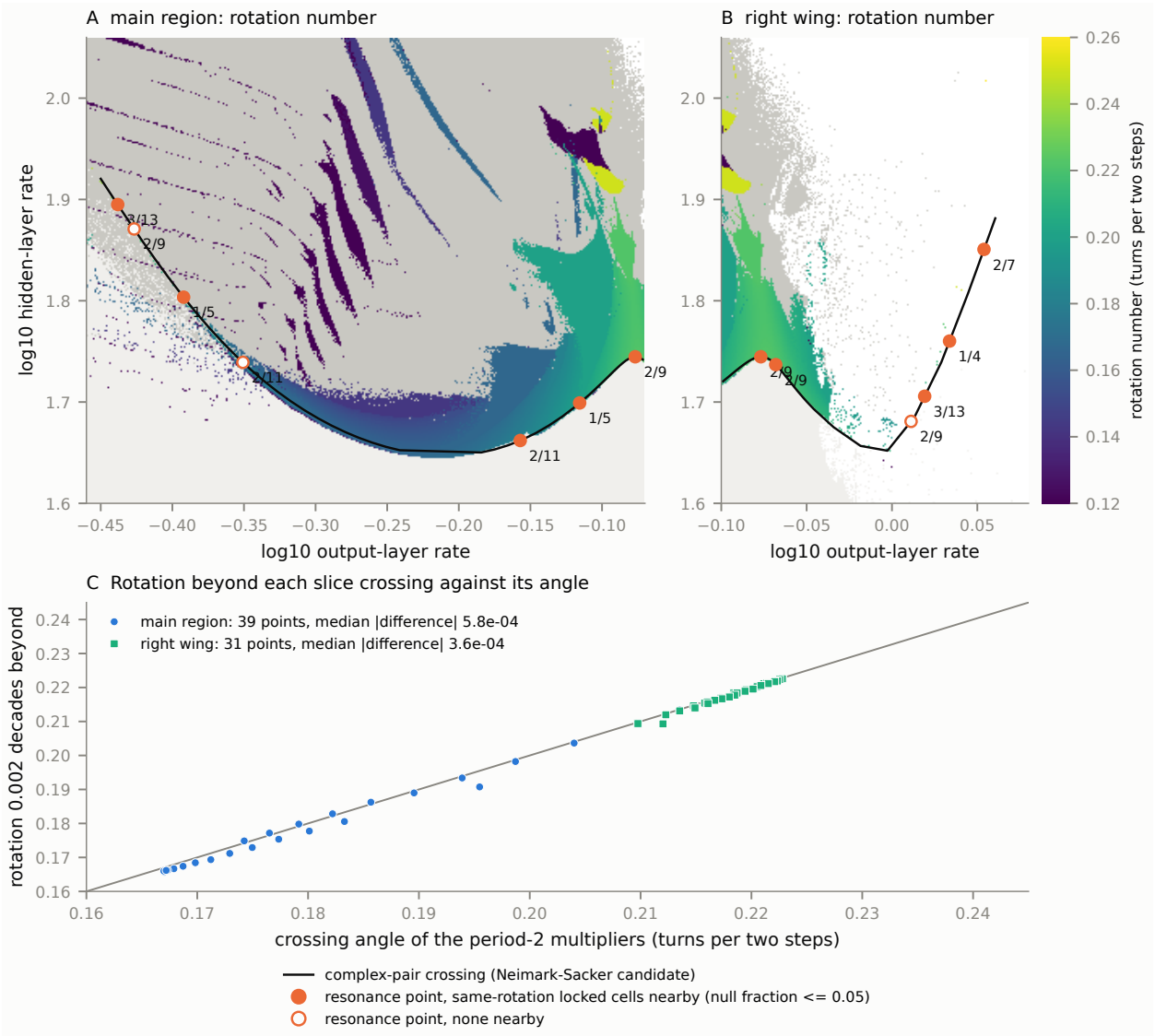


Figure 2: Beyond slice-wise period-two crossings, measured rotation tracks the crossing angle and locked windows appear as plateaus. A and B: rotation-number maps of S2-28011 (1,024 × 1,024 and 1,024 × 400 cells) with the merged crossing locus (69 points, black guide between samples) and provisional resonance locations where the interpolated crossing angle is m/q (filled when locked cells of the same rotation lie closer to the point than in 95% of random placements, hollow otherwise; the 95% level is a descriptive cut); chaotic cells gray, divergent white, period 2 light. C: rotation 0.002 decades beyond the curve against the crossing angle at the same output rate. Resonance points are interpolated between output-ordered slice results; their full-state branch association is unverified. *Generated by* `paper/gd-route/figures/f2_rotation_map.py`; *receipt* `figures/f2_rotation_map.provenance.json`.

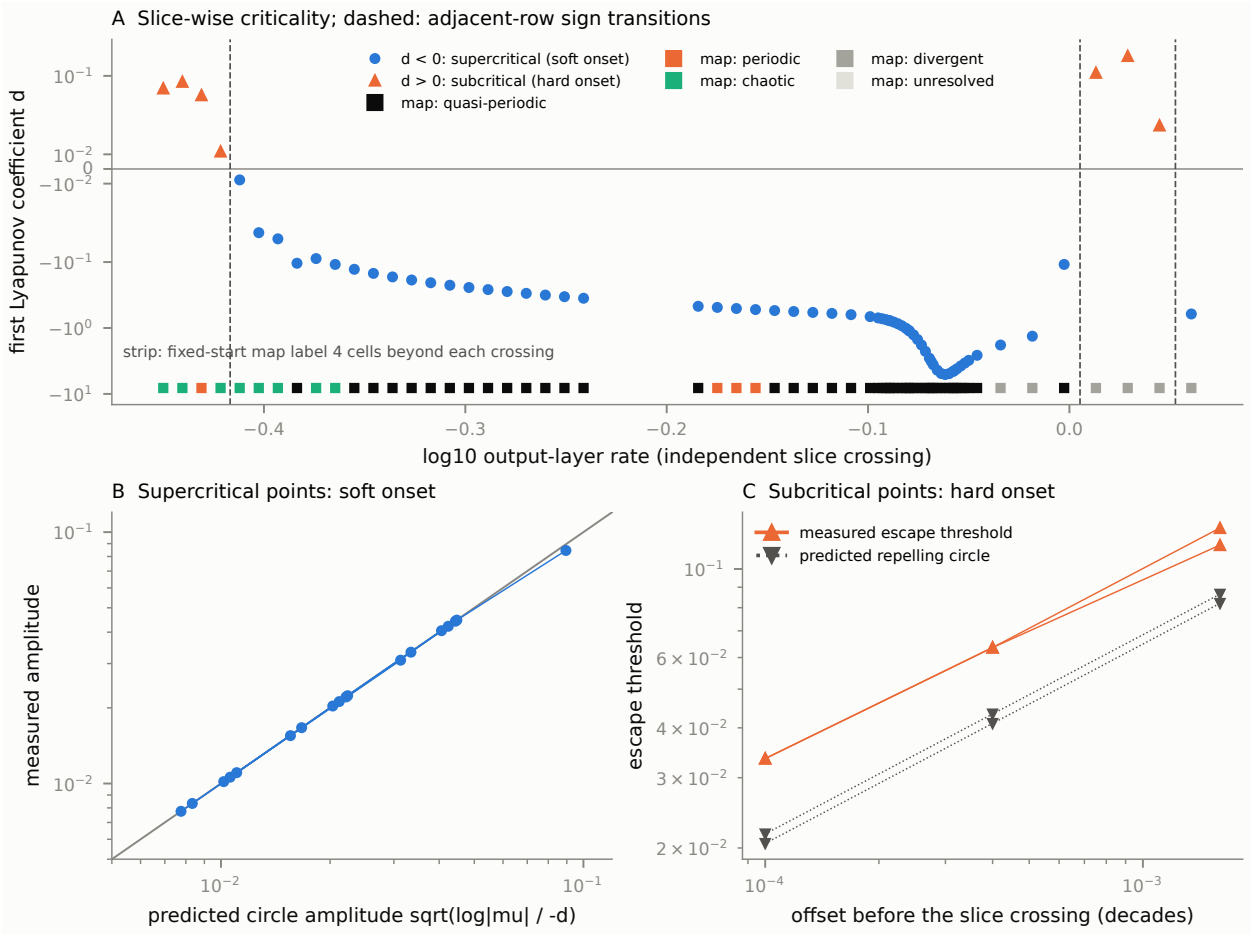


Figure 3: The first Lyapunov coefficient is negative at most sampled crossings and predicts local onset at the tested points. A: d at the merged slice-wise crossings (symmetric-log axis); dashed lines mark sign transitions between adjacent output-ordered rows. Branch identity across these rows is unverified, so these transitions do not locate sign changes along one established orbit branch. The strip shows the label of the fixed-start rotation maps four cells beyond the curve. B: at supercritical points, the amplitude of the circle reached from near the 2-cycle against the normal-form prediction, at three offsets beyond the curve. C: at subcritical points, the escape threshold before the curve against the offset, with the predicted repelling circle. A point counts as consistent when its amplitudes match within 6% (supercritical), or when the state jumps to an offset-independent distant amplitude and the escape threshold grows like the square root of the offset (subcritical). *Generated by* `paper/gd-route/figures/f7_criticality.py`; *receipt* `figures/f7_criticality.provenance.json`.

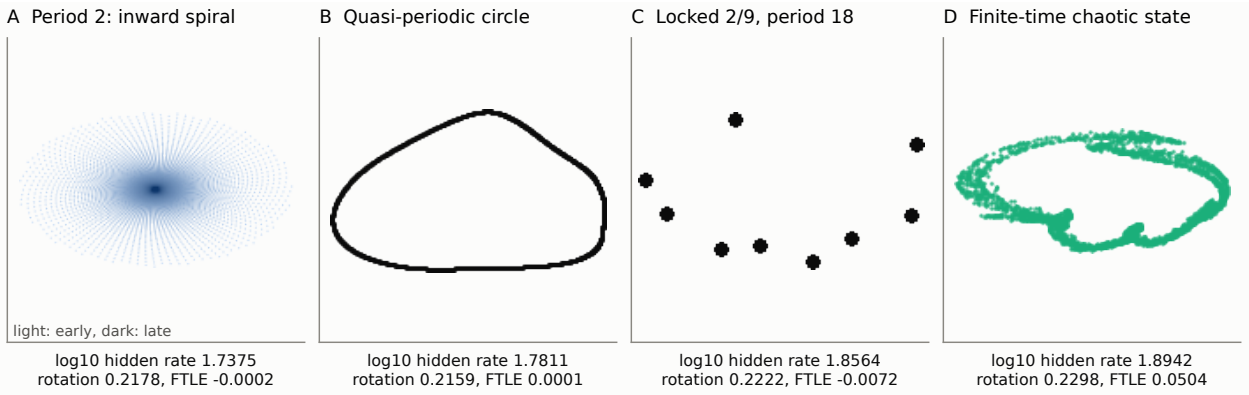


Figure 4: Four states along one slice, in order of increasing hidden-layer rate: an inward spiral to the 2-cycle, a closed projected curve, nine locked points and an irregular projected state. Each panel shows every second parameter vector of one run projected on its own two principal axes. The rates are chosen by a fixed rule from the staircase receipt (the period-2 cell closest to the onset, the median quasi-periodic cell, the middle of the longest 2/9 plateau, the median chaotic cell between the 2/9 and 1/4 plateaus), not by eye. The states are not a monotone sequence: beyond the onset the rotation number first falls, and beyond panel D training locks again at 1/4 before chaos returns (Figure 5). Projections onto two axes can hide structure in the remaining dimensions. *Generated by* `paper/gd-route/figures/f6_route_portraits.py`; *receipt* `figures/f6_route_portraits.provenance.json`.

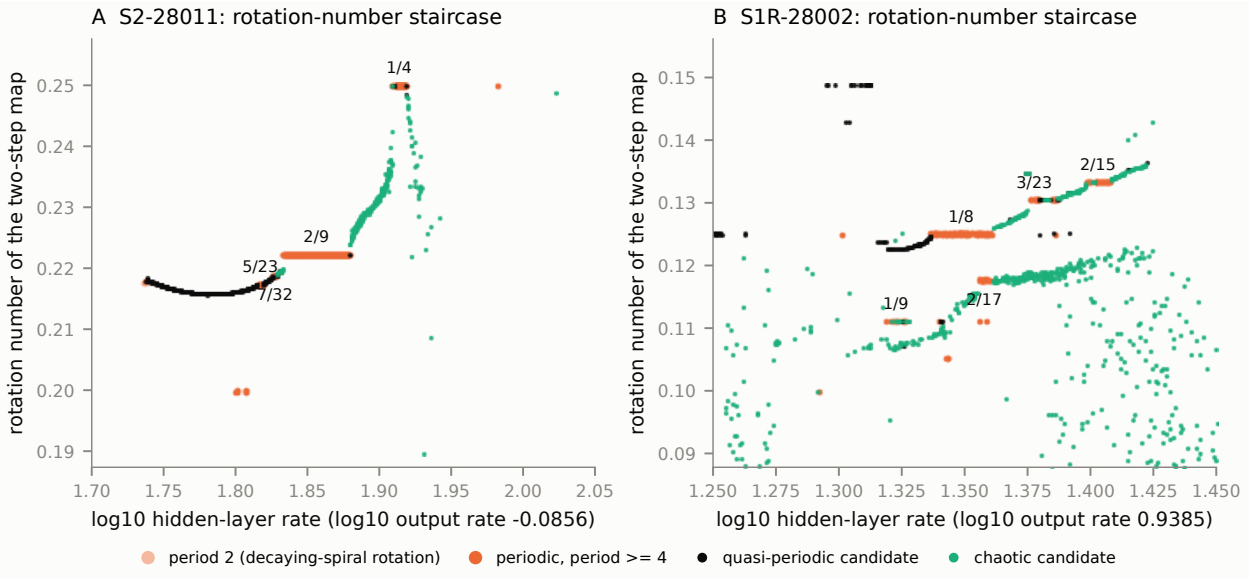


Figure 5: Along slices, quasi-periodic cells trace continuous rotation numbers and periodic cells sit on rational plateaus. Plateaus are detected automatically (locked cells with period $2q$ or a doubling of it, rotation within 6×10^{-4} of m/q) and labeled when they hold at least 8 cells. In S2 the rotation number first falls after the onset, then rises; the cells between the 2/9 and 1/4 plateaus are almost all chaotic candidates. In S1R, rotation bands occupy overlapping rate ranges; this projected scan alone does not establish coexisting full-state attractors at common parameters. Chaotic cells outside the plotted range are not shown. *Generated by* `paper/gd-route/figures/f3_staircases.py`; *receipt* `figures/f3_staircases.provenance.json`.

6.4 Rotation numbers form staircases

Along the slice through the period-18 window of S2-28011 (Figure 5), the rotation number falls from 0.2180 at the onset to 0.2158 at $\log_{10}\eta_h = 1.780$, then rises and locks at 2/9 for $\log_{10}\eta_h$ from 1.833 to 1.879. Of the 168 cells between the 2/9 and 1/4 plateaus, 167 are chaotic candidates; training then locks at 1/4 from 1.909 to 1.919 before chaos returns. The labeled plateaus are 5/23, 7/32, 2/9 and 1/4 in S2 and 1/9, 2/17, 1/8, 3/23 and 2/15 in S1R. Consecutive plateaus are Farey neighbors ($|m_1q_2 - m_2q_1| = 1$) in 3 of 3 pairs in S2, against 49% per pair for randomly drawn fractions in the same range, and in 4 of 4 pairs in S1R, where the null is 100% because every fraction with a small enough denominator in that range appears. The Farey pattern is weak evidence in S2 and uninformative in S1R.

6.5 A Birkhoff ordering test of the windows

Across the 61 windows, the 3,904 rays end at 1,316 folds, 1,012 flips and 1,395 complex-pair crossings (181 rays reached the end of their range or stayed unresolved; Figure 6A). Fold edges are what the edges of a locked window on a circle look like, while flips and secondary complex-pair crossings are expected where a locked orbit loses stability off the circle.

The Birkhoff test results depend on which orbits are eligible, so we give each denominator. Of the 129 numerically validated orbits with period at least 3, 115 have at least 5 two-step points; 65 of these pass the plane test, against 1.57 expected by chance. All 61 confirmed windows are eligible for the plane test and 31 pass, against 4.92 expected; this count includes 4 windows with at most 3 points, which pass by construction. For the refined test, 56 confirmed windows have at least one test on 5 or more points (the 5 excluded windows have periods 4, 4, 6, 6 and 8); 35 of them pass, against a union bound of 3.58 chance passes. 28 pass at the two-step level, with measured shifts $m = 1$ for 19, $m = 2$ for 8 and $m = 3$ for 1; 7 pass first at a period-doubled level, for example period-20 orbits ordered under four steps with rotation 2/5, twice 1/5. Of the 5 odd-period windows, 0 pass (null 0.17): they did not pass this diagnostic, and their organization is open.

6.6 Resonance ancestry of a rule-selected window: not established

The test asks whether the rule-selected 1/5 window of S2-28011 descends from the 1/5 resonance of the crossing curve. The controls behaved as required: a resonant normal form's 1/5 tongue passed, the same branch scored against a displaced root was refused (alternative ancestry), and a logistic period-5 window born by a saddle-node was refused (alternative ancestry). At the right 1/5 resonance point 6 of 6 onset criteria passed (complex pair, other multipliers inside, transverse crossing, attracting circle with radius exponent 0.503, converging invariance residual, rotation equal to the crossing angle); at the left point 5 of 6 passed, the radius exponent 0.829 lying outside the frozen band.

From the right point, the locked interval has two fold edges on all 6 cross-sections, and its width and amplitude shrink toward the point (Figure 7). The frozen estimator nevertheless extrapolates the center to 1.5×10^{-4} decades from the predicted point, 53% above the tolerance of 10^{-4}, so the frozen outcome is alternative ancestry. The connection from the tracked interval to the window ended in exhausted budget, and the left point gave no qualifying window. Under the frozen rules, the ancestry of this window is not established. Post hoc, and not replacing that outcome: the two shallowest levels lie outside the asymptotic regime, and over the three deepest levels the width scales with exponent 1.518 and the amplitude with 0.513, against $(q-2)/2 = 1.5$ and 0.5 for a Neimark–Sacker resonance tongue with $q = 5$.

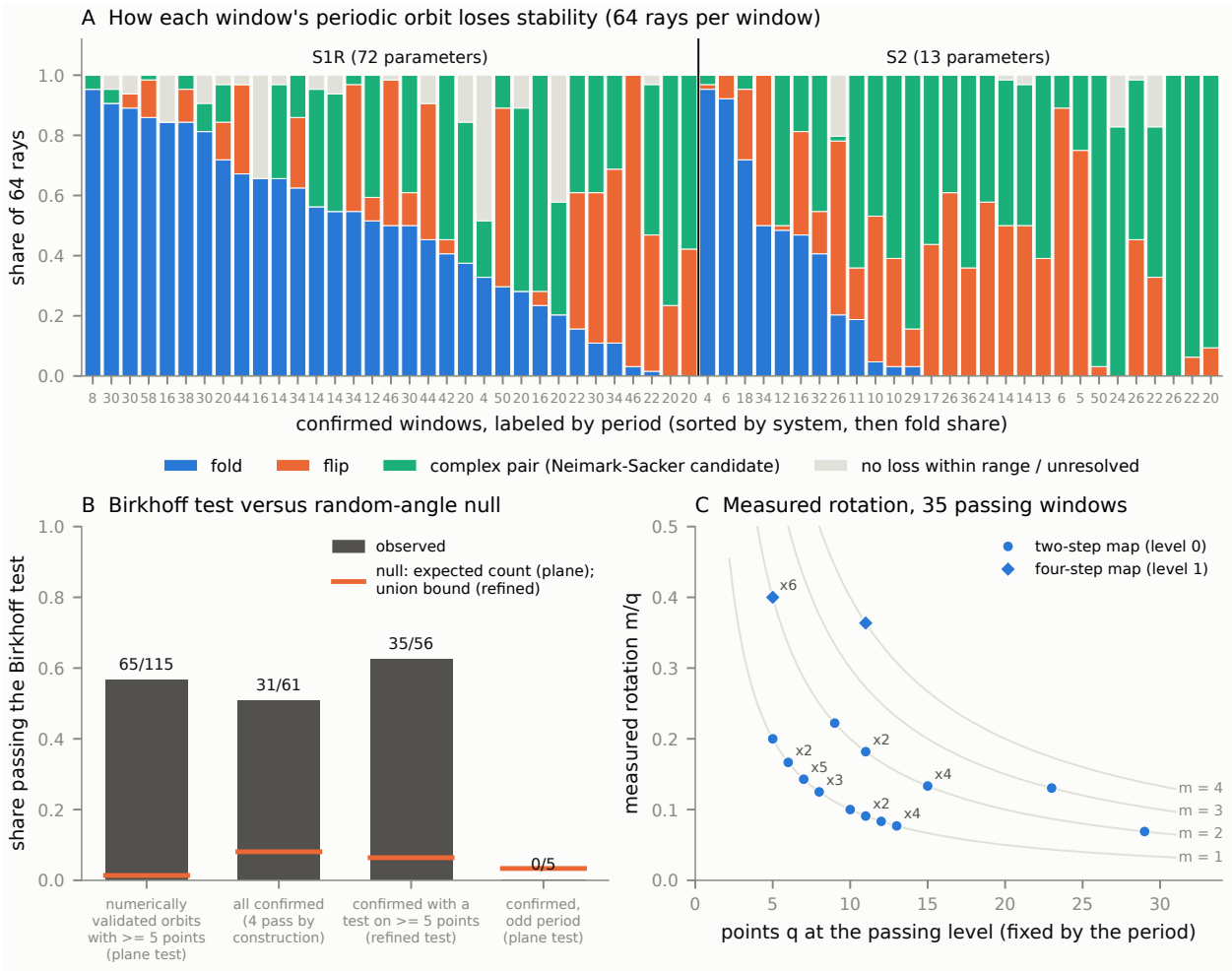


Figure 6: Across all 61 windows, stability boundaries mix fold, flip and complex-pair crossings, and most eligible windows pass a Birkhoff ordering test far above its null. A: boundary type along 64 rays per window. B: share of orbits passing the test (gray) against the null (orange): the expected share of chance passes for the plane test, and the union bound for the refined test; eligibility is printed under each bar. C: rotation m/q measured by the passing test at its first passing level; q is fixed by the period and level by construction, so only the shift m and the ordering are measured. *Generated by* `paper/gd-route/figures/f5_window_census.py`; *receipt* `figures/f5_window_census.provenance.json`.

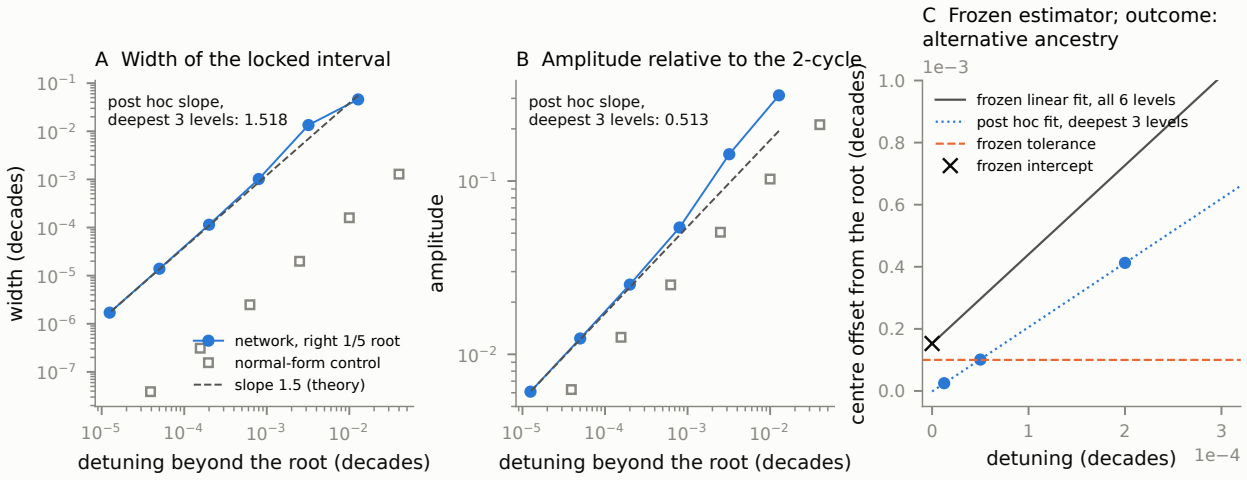


Figure 7: Near the predicted 1/5 resonance point, a fold-bounded locked region narrows with the scaling of a resonance tongue, but the frozen estimator places its tip outside the tolerance. A and B: width and amplitude of the locked period-10 interval on six cross-sections against the distance beyond the resonance point, with the resonant normal-form control and the theory slopes; the slopes printed are post hoc fits over the three deepest levels. C: the frozen estimator (linear fit of the interval's center over all six levels) and its intercept against the frozen tolerance, with a post-hoc fit over the deepest three levels; the axes show the deepest three levels. *Generated by* `paper/gd-route/figures/f8_ancestry.py`; *receipt* `figures/f8_ancestry.provenance.json`.

6.7 Classical shrimp anatomy is the exception

Draft criteria for classical shrimp anatomy require fold and flip boundaries each on at least 8 of 64 rays, at least half of the flip rays opening onto period $2p$ in a local map, and agreement between ray and map edges on at least three quarters of rays. 15 windows meet the boundary-count part and 2 meet all of it. The clearest, the period-18 window of S2-28011 (Figure 8), has 46 fold, 15 flip and 3 complex-pair rays; ray and map edges agree within two cells on 61 of 64 rays, and 10 of its flip rays open onto period 36. Complex-pair boundaries, outside the fold–flip anatomy used by our draft criteria, are common: 41 of 61 windows have at least 8 such rays.

6.8 Beyond these two networks

Distinct continuation outcomes in S1R. In the 72-parameter S1R network, two bootstrap procedures give different continuation outcomes. Of 36 output rates traced from one start, 16 end in a complex-pair crossing and 20 starts did not pass the finite-horizon period-two tail screen; from a lower start, 25 of 36 end in a complex-pair crossing, 4 in folds, and 7 starts did not pass that screen. The procedures can reach different crossing locations and angles, but their historical summaries do not retain states for a common-parameter full-state comparison. A subsequent saved-map audit keeps these starting families separate: each output-rate slice was initialized independently, and adjacent rows need not represent one continuous orbit branch. The retained near-curve map cells remain unresolved under the original status classifier. Provisional rational-angle intersections do not establish association with distant same-period windows; saved full-state continuation is still required. See `docs/explore-s1r-rotation-audit-receipt.json` and its separate replay review, `docs/reviews/s1r-rotation-map.md`.

After a small hidden-rate backoff, a new full-state pilot follows one saved criticality lineage to 5 nearby output rates at a fixed hidden rate. Each neighboring link passes direct, two-half-step

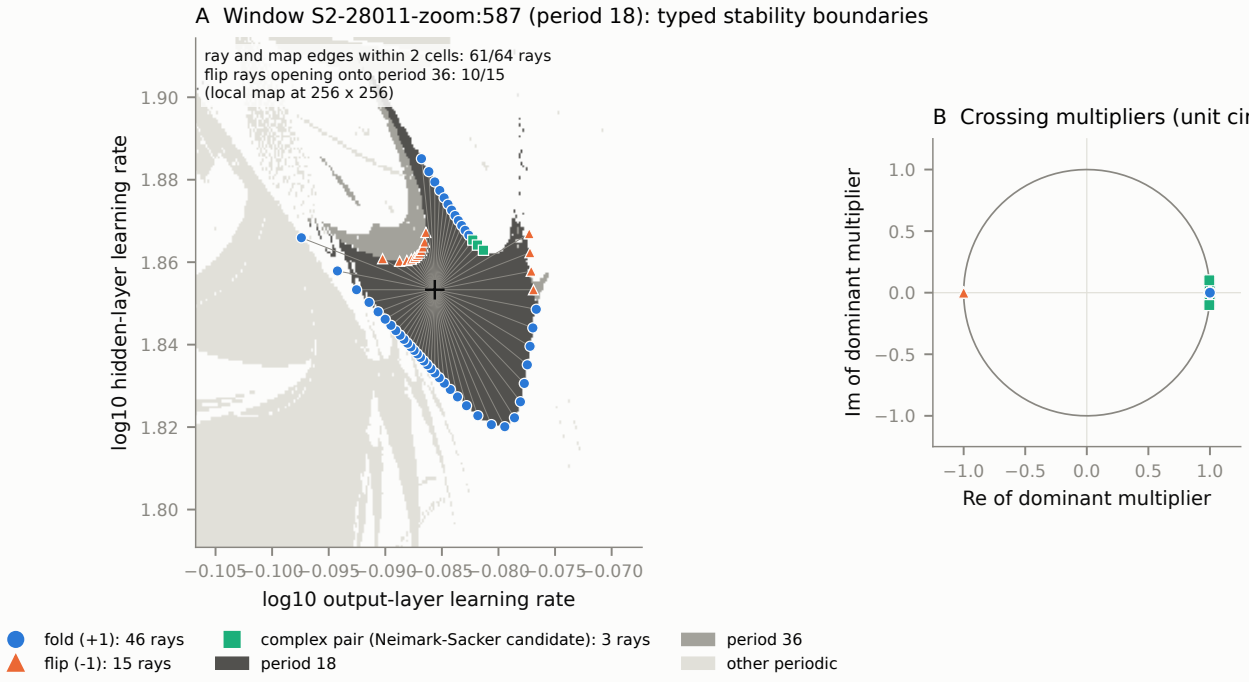


Figure 8: The period-18 window of S2-28011 has a fold-bounded body with flip edges opening onto period-36 bands. A: local map from the shared initial parameters with the window's orbit continued along 64 rays; markers show where and how it loses stability. B: crossing multipliers on the unit circle. This window was selected as the clearest example and is one of only 2 meeting the draft shrimp criteria; it is not representative (Figure 6). *Generated by* `paper/gd-route/figures/f4_window_anatomy.py`; *receipt* `figures/f4_window_anatomy.provenance.json`.

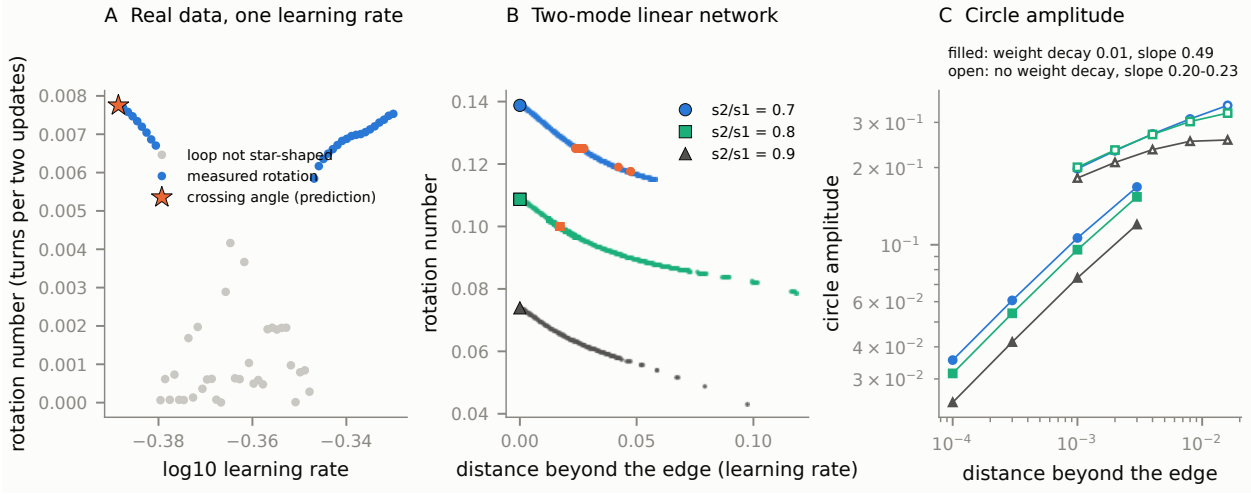


Figure 9: The same crossing, with the rotation predicted by its angle, in a real-data network with one learning rate and in Hofmann's two-mode linear network, where symmetry-reduced observables show soft onset and locking beyond the edge. A: rotation number on the circle beyond the crossing of a 529-parameter tanh network on digit images (gray where the loop is not star-shaped about its centroid, so the angle estimate is invalid), and the crossing angle (star). B: rotation number against the distance beyond the rotational edge for three singular-value ratios, with weight decay; locked cells in orange. C: circle amplitude against the distance beyond the edge with and without weight decay; exponents are slopes between the two smallest offsets. *Generated by* `paper/gd-route/figures/f9_generality.py`; *receipt* `figures/f9_generality.provenance.json`.

and reverse consistency checks, with both phases independently corrected and primitive period two numerically validated. All primary points are numerically attracting; their two-step spectral radii range from 0.998661 to 0.998991. An independent audit replays the saved graph and reconstructs full analytic gradient/Hessian monodromies at the anchor and all primary points. This establishes local numerical consistency at the tested steps, not branch uniqueness or a continuous parameter enclosure. That local pilot alone does not associate a bootstrap family, establish coexistence at common controls, or connect a distant window. See `docs/s1r-branch-association-receipt.json` and `docs/reviews/s1r-branch-outcome.md`.

A subsequent bounded run reproduces both bootstrap procedures with complete saved provenance, using 40,000 updates per start at log hidden rates 1.10 and 0.95 and the same output rate. Refinement of both tails yields numerically attracting primitive period-two pairs at their respective controls. Following the saved lineage through 12 hidden-rate links reaches exactly the upper bootstrap's controls. Every direct, two-half-step and reverse witness passes; after cyclic phase exchange, the normalized full-pair discrepancy is 2.65×10^{-15}. An independent audit checks all recorded qualifications and link decisions, with separate analytic derivatives at fixed representative pairs and direct reevaluation of saved tail transitions. It checks counters and checkpoints for the full warmups, without replaying unsaved intermediate updates. This associates the saved lineage with the new upper-start reproduction at the tested controls. It does not recover the missing historical states, identify the lower-start family at common controls, prove coexistence, or connect a distant window. See `docs/s1r-bootstrap-association-receipt.json` and `docs/reviews/s1r-bootstrap-outcome.md`.

A separate attempt to reach the lower-start controls completes 54 of 75 planned macro steps, comprising 56 accepted witnessed links. The last connected pair is at $\log_{10} \eta_h = 0.992$ and is numerically unstable. The next macro target is 0.990; its first phase corrector stalls at the initial

trial and all 4 permitted step halvings. The run retains this gap and makes no lower-endpoint comparison. An independent audit checks the saved path and failure history, including separate analytic derivatives at fixed representative pairs. This leaves the lower-family association unresolved; it establishes neither a fold, global disconnection, coexistence nor absence of a connecting branch. See `docs/reviews/s1r-lower-outcome.md` and `docs/reviews/s1r-lower-independent-outcome.md`.

A fixed-state follow-up finds that the stored direction solves its local Newton equation accurately. With $R = F^2(\theta) - \theta$ and $A = D_\theta R$ at fixed rates, the stored direction has $\|Ap + R\|_2/\|R\|_2 = 3.138 \times 10^{-9}$ and Frobenius-normwise backward error 1.63×10^{-16}. Nevertheless, all 13 retained line-search trials increase the original unscaled infinity-norm residual; the smallest ratio is 1.33778. Independent saved-data checks reproduce the equation and trial reductions using retained derivatives. The qualified predecessor and last accepted pairs remain numerically regular under the declared rank test. The terminal failed state is not a closed orbit: its raw residual Jacobian has condition estimate 1.46×10^7, which does not establish a fold. At that nonclosed state, raw- and scaled-metric secant conventions choose opposite orientations of the local null direction; its sign therefore does not establish a branch turn. That diagnostic performs no correction or continuation. See `docs/reviews/s1r-lower-gap-outcome.md` and `docs/reviews/s1r-lower-gap-independent-outcome.md`.

A subsequent pseudo-arclength pilot completes 2 locally witnessed links from the last qualified pair. All 9 qualified period-two pairs, including 6 auxiliary witnesses, remain numerically unstable. Each link passes direct, two-half-step and reverse checks, including both native phases and state comparisons at identical controls. With $h = \log_{10} \eta_h$, a fixed raw Euclidean metric on (θ, h) and consistent carried-phase orientation, the second link runs from $(h, t_h) = (0.99190473, -0.014124)$ to $(0.99189083, +0.009173)$, where t_h is the hidden-rate component of the unit tangent. Its qualified midpoint witness has $(h, t_h) = (0.99188533, -0.003697)$. This supports a local numerical turning candidate on the unstable cycle segment, not an attracting-window boundary. No singular point, simple $+1$ multiplier, parameter transversality or quadratic nondegeneracy is established, so it is not a fold proof. The one-thread producer and independent saved-data audit take 5.20 and 1.50 seconds, respectively; the audit reuses saved derivatives and decompositions. The combined change of chart, predictor and globalization does not isolate which change helped, and the lower-family connection and any training utility remain unresolved. See `docs/reviews/s1r-pseudoarclength-outcome.md` and `docs/reviews/s1r-pseudoarclength-independent-outcome.md`.

Real data, one learning rate. A 64-8-1 tanh network (529 parameters) trained with one global learning rate and weight decay on 1,024 standardized UCI digit images (two-class task) has a numerically validated period-2 orbit that loses stability at $\log_{10} \eta = -0.3885$ with dominant two-step multiplier $0.998814 + 0.048693\mathrm{i}$, a complex pair on the unit circle. Its angle predicts the rotation of the circle beyond: 0.0077528 turns per two updates, against 0.007751 measured just beyond the crossing on a smooth projected loop consistent with an invariant circle (FTLE -0.0006; Figure 9A). This is one branch among those examined: of 4 numerically validated period-2 branches continued to their loss of stability, 1 ends in a complex-pair crossing and 3 yield boundary candidates with a real multiplier near $+1$, consistent with folds. On a second data subset Newton did not resolve a period-2 orbit to the stated tolerance from 2 starts. No locked window was resolved; at rotations near 0.008 the relevant tongues have very large denominators.

Oscillatory onset beyond the rotational edge of a linear network. In the two-mode depth-two linear network of Hofmann [15], the aligned 2-cycle loses stability at his rotational edge $a(s_1 + s_2) = 2$, which we reproduce to 2.2×10^{-16} for $s_2/s_1 = 0.7, 0.8, 0.9$, with crossing angles

0.1380, 0.1082 and 0.0737 and a transverse crossing ($d|\mu|/da = 1.85$–1.95). Beyond the edge, rotation-invariant observables show an attracting closed curve of the two-step dynamics. An exact rotation symmetry leaves a neutral multiplier at one, including with weight decay, so the generic isolated-orbit Neimark–Sacker theorem requires symmetry reduction here. Without weight decay the second mode's minima form a neutral continuum and the circle amplitude grows with exponent 0.20–0.23, a degenerate bifurcation; with weight decay 0.01 the exponent is 0.49, the square-root growth consistent with supercritical onset in the symmetry-reduced dynamics, and the rotation number at an offset of 10^{-4} is within 3.3×10^{-5} of the crossing angle (Figure 9B and C). Beyond the edge the rotation locks at 1/8 (period 16), 2/17 (period 34), 5/42 (period 84) for $s_2/s_1 = 0.7$, and at 1/10 (period 20) in a fine scan at 0.8. The same crossing occurs in the two-weight scalar factorization $\frac{1}{2}(uv - d)^2 + \frac{1}{2}\lambda(u^2 + v^2)$ with $\lambda = 0.2$ and unequal layer rates (crossing angle 0.4288; the circle amplitude grows by a factor 2.08 for a fourfold offset, close to the square-root value 2), found as one of 6 of 64 cases of a coupled four-weight family that ended in a complex-pair crossing; boundary types for two held-out couplings were predicted correctly in 32 of 32 cases.

A scalar mechanism with an explicit equal-rate control. For $L(u, v) = \frac{1}{2}(uv-d)^2+\frac{\lambda}{2}(u^2+v^2)$, $0 < \lambda < d$, equal rates preserve $u = v = x$. Set $c = 1 + \eta(d - \lambda)$ and $e = 1 - \eta(d + \lambda)$. The positive balanced 2-cycle for $c > 2$ is

$$x_\pm = \frac{\sqrt{c+2} \pm \sqrt{c-2}}{2\sqrt{\eta}}, \qquad m_\parallel = 9 - 2c^2, \qquad m_\perp = 1 + e^2 + ec. \tag{8}$$

The rescaled cubic cycle and longitudinal multiplier recover the balanced scalar formulas in Hofmann [15, Propositions 2–3]; the transverse multiplier tests the full two-weight state. Both multipliers are real, so this branch cannot have an NS crossing; its longitudinal flip is at $\eta = (\sqrt{5} - 1)/(d - \lambda)$ when the transverse direction remains attracting. Unequal rates can break the common eigendirections of the symmetric step factors, allowing complex two-step multipliers. This explains a branch-specific mechanism and does not exclude other equal-rate attractors.

A fixed-target exploratory scan over ridge and rate ratio gives 12 flips and 5 complex-pair crossings in 17 followed branches; all sampled equal-rate cases flip. All complex crossings have negative computed criticality. In the representative case the crossing rate is 0.733508896, birth rotation 0.428819 and coefficient −3.836452; three decreasing detunings give an amplitude exponent of 0.49790. Analytic composed-map derivatives agree with an independent finite-difference calculation. These are numerical checks of local onset, not a complete phase diagram or a rigorous invariant-circle theorem for that numerical representative. Section 5.2 gives a separate local proof for another selected scalar case. The compact receipt is `docs/explore-minimal-model-scalar-receipt.json`.

6.9 Consequences for training

The regime matters for training, but what it predicts is largely generic (Table 3). On a slice of 750 cells through the period-18 window, perturbations of the initial parameters die out on period-2 and most locked orbits, persist in proportion to their size on invariant circles, and saturate in chaos; locked windows reached through chaotic transients split into runs that reproduce exactly and runs that land on another phase of the same cycle. These differences follow the sign of the Lyapunov exponent, transient amplification, oscillation amplitude, sharpness and rate. Adding locking and rotation features to a model of those quantities gives an incremental R^2 whose interval lies above zero for 1 of 7 outcomes (the spread over anneal start times). In an analysis pooled over two slices with slice-specific slopes, changed after seeing an earlier fit and therefore adapted, two locking effects remain: the dose-response of disagreement on the perturbation size ($\Delta R^2 = +0.179$, [+0.070,

Outcome (median per regime)	period 2 (282 cells)	quasi-periodic (125 cells)	locked (97 cells)	chaotic (223 cells)	ΔR^2 [95%]
function disagreement, 40k updates, start 10^{-8} apart ($\log_{10}$)	−15.87	−6.11	−14.69	−0.63	+0.000 [−0.179, +0.067]
function disagreement after the step-decay anneal ($\log_{10}$)	−15.81	−8.70	−9.41	−2.34	−0.005 [−0.243, +0.070]
spread over 64 anneal start times ($\log_{10}$)	−4.98	−3.73	−3.24	−2.36	+0.045 [+0.001, +0.107]
training loss, last 8,192 updates ($\log_{10}$)	−2.20	−2.10	−2.04	−1.93	−0.001 [−0.010, +0.001]
held-out loss, last 8,192 updates ($\log_{10}$)	−2.06	−1.94	−1.86	−1.73	−0.001 [−0.007, +0.001]
held-out loss after the anneal ($\log_{10}$)	−2.98	−2.98	−2.99	−3.05	+0.057 [−0.169, +0.232]
loss power in spectral lines	1.00	0.99	0.96	0.10	+0.001 [−0.037, +0.010]

Table 3: Consequences per regime (exploratory, S2-28011, one slice of 750 cells). Columns give medians per regime (cell counts in parentheses). ΔR^2 is the cross-validated gain from adding locking and rotation features (locked and quasi-periodic indicators, rotation number, distance to the nearest rational, $1/q$) to a base model of FTLE, sharpness, rate and oscillation amplitude, over the 729 cells beyond the edge of stability, with moving-block bootstrap intervals. Regime differences are large; the incremental value of rotation information is not.

+0.357]) and the spread over anneal start times (+0.047, [+0.011, +0.099]); both are carried by the locked indicator rather than the rotation number.

Controlled rate ramps separate criticality from loss spikes. At two previously inspected S2 points, one with positive and one with negative criticality, we match stable and unstable endpoints in $\log|\mu|$, use two ramp speeds, and fork identical terminal states into unchanged rates, reversing the hidden-rate ramp, or halving both rates. For near-orbit starts, the hard point leaves the local branch and does not return under the small reverse ramp; the soft point stays nearby and returns. Nevertheless, both spike: postcrossing peak loss relative to the instantaneous frozen-cycle maximum is 9.59–11.23 at the hard point and 1.60–1.62 at the soft point. A loss spike alone therefore does not identify hard criticality. Halving both rates gives terminal mean losses of 0.570 and 0.494 times the original cycle means. That generic baseline performs better on this endpoint, although its abrupt change can itself produce a transient peak. The interventions have different sizes; this is not evidence of a superior geometry-based controller. Ordinary initialization at the hard point misses the local branch before crossing, so those runs are basin-accessibility controls. Selection and thresholds are exploratory, and the result covers only these two points in one network. Full-state replay and the post-run basin annotation are bound by `docs/warmup-intervention-receipt.json`.

Intervention-dose feasibility (negative). Before collecting further intervention outcomes, we checked a candidate comparison at 4 saved ramp-terminal states. Let g be the gradient and P the baseline block learning-rate matrix. A uniform fractional reduction $\delta = 0.05$ changes the update by $R = \delta\|Pg\|$ relative to the unchanged-rate sham. Matching that displacement using only block b requires fractional reduction $r_b = R/\|(Pg)_b\|$. At the four states, the required output-block fractions are 1.412, 1.444, 0.766, 0.766, all above the declared cap 0.25. This fixed candidate design is infeasible; the first two fractions would even reverse the output update. No pulse or training update was applied. An independent analytic-gradient implementation reproduces the decision. Equal

sham-relative displacement would still leave total step lengths unequal, and comparable distant periodic and chaos-qualified controls remain necessary for a geometry-specific claim. The result rejects this dose definition at these states, not geometry-based intervention in general. See `docs/reviews/utility-dose-outcome.md` and `docs/reviews/utility-dose-independent-outcome.md`.

Reading the rotation from the loss. The dominant peak of the even-step training-loss spectrum equals the two-step rotation number within 10^{-3} in 125 of 125 quasi-periodic and 97 of 97 locked cells of the S2 slice, and in 31 of 31 and 19 of 19 on the S1R slice. Before the crossing, with small noise added to every update, an autoregressive fit of the even-step loss recovers the angle of the period-2 orbit's complex multiplier with median error 1.9×10^{-4} over 41 period-2 cells of S2 (largest 9.4×10^{-4}) and 3.6×10^{-4} over 34 cells of S1R; the estimator was chosen after a first pass (adapted). We treat this as local system identification, not early warning, until sampling, harmonic, abstention, autoregressive-baseline and specificity rules are fixed in advance.

Placeholder. Prespecified route-to-chaos path: a connected path, fixed in advance, from the period-2 region through the quasi-periodic band and locked windows into sustained chaos with converged diagnostics (not yet run).

CIFAR development pilot (unresolved). The development-subset learning-rate sweep has completed and shows a slowly drifting period-2 candidate followed by loss modulation. The initial full-state Newton–GMRES continuation was no longer running after its configured time limit and left no saved orbit result. A reviewed revision adds incremental checkpoints, convergence checks at every returned state and residual-qualified matrix-free multiplier estimates. A bounded correction then failed its linear-solver tolerance before any Newton update; the saved state remains outside the closure gate and no spectrum was computed. A diagnostic at the same saved state and Krylov budget gives a true linear forcing ratio 0.8843, above the declared 0.5 limit. Two of four fixed damping trials reduce the residual merit, but none passes the forcing or full-state closure gate; the smallest relative closure residual is 4.86×10^{-6}, against 10^{-8} required. An independent saved-array audit reproduces these decisions; it does not independently reevaluate the model's saved map images or Jacobian products. This diagnoses a failed correction attempt, not absence of a period-two orbit. See `docs/cifar-linear-diagnostic-receipt.json` and `docs/reviews/cifar-linear-diagnostic-outcome-review.md`.

A later matched comparison replaces 4 restart-50 cycles with one restart-200 cycle, retaining 200 inner iterations at the same saved state. Its first 50 callback residuals reproduce the saved baseline. The true linear forcing improves from 0.8843 to 0.6620, but remains above 0.5. All 4 fixed trials now increase the absolute residual merit; none passes Armijo, closure or the candidate-step gates, and no update is applied. An independent saved-data audit reproduces these decisions without regenerating map images or Jacobian products. This single-state comparison does not establish absence of a periodic orbit or a general solver advantage. See `docs/reviews/cifar-restart-outcome.md` and `docs/reviews/cifar-restart-independent-outcome.md`. Modulated loss alone does not establish an invariant circle or a Neimark–Sacker crossing. The second subset remains reserved for a separately frozen test; no result from it is included here.

An exact structural reduction, with a negative solver diagnostic. For the same fixed-input full-batch model with ridge term $\lambda\|\theta\|^2/2$, write the first-layer weights as $W = QB + N$, where Q has orthonormal columns spanning a space containing range$(X^\top)$, $Q^\top N = 0$, and X contains the fixed training inputs as rows. The data loss is independent of N, so plain GD with common scalar learning rate η separates as $(z, N) \mapsto (G(z), qN)$, with $q = 1 - \eta\lambda$ and z containing B and

the other parameters. If $q^p \neq 1$, every period-p orbit has $N = 0$; its primitive period is preserved in the reduced map and its monodromy is the reduced monodromy direct-summed with $q^p I$. With a nonzero complementary subspace, an attracting reduced orbit attracts in the full map exactly when $|q^p| < 1$. These statements do not apply unchanged to adaptive updates, changing input spaces or nonisotropic regularization.

Keeping all 512 columns of the thin QR factor, without rank truncation, reduces the nontrivial coordinates from 98,666 to 16,746 in this development case. However, the complementary directions contain only 1.75×10^{-8} of the original squared period-two residual, below the fixed 0.5 priority criterion. A saved-data audit independently reproduces the projections and decay identities; it does not regenerate the original map images or Jacobian products. Thus this decomposition does not explain most of the observed residual or establish a convergence or speed advantage. Removing N can also change predictions on inputs outside the training span. See `docs/plans/cifar-input-span-diagnostic.md`, `docs/reviews/cifar-input-span-outcome.md` and `docs/reviews/cifar-input-span-independent-outcome.md`.

Fixed curvature approximation (negative exploratory pilot). All 6 selected period-two references in S2-28011 and S1R-28002 pass the unchanged full-state and historical-spectrum checks. Nevertheless, approximating each whitened phase Jacobian by its top-curvature direction plus an isotropic bulk fails to predict the leading pair of the native derivative $DF(F(\theta))DF(\theta)$. Unordered pair errors, normalized by $\max(1, \rho(M))$, range from 0.0802 to 1.9349; every reference exceeds the 0.02 gate. The 2 real controls are falsely predicted complex, while the stable-complex S1R reference is predicted real. The mean error 0.8586 is below the 1.2023 mean of the better aligned or orthogonal commuting control at each reference, but this relative gain does not rescue the absolute errors or class mistakes. This rejects the chosen rank-one/bulk approximation on this fixed panel, not richer curvature mechanisms; it establishes no larger-model or language-model utility. See `docs/plans/curvature-subspace-mechanism.md` and `docs/reviews/curvature-six-reference-outcome.md`.

Actual-text transformer pilot (architectural feasibility). A one-layer causal transformer with 15,776 parameters, tied token/output embeddings and LayerNorm was trained on 256 fixed next-character targets per update from tiny-Shakespeare. Full-batch GD with ridge regularization makes the update autonomous. Of 10 exploratory rates from one shared initialization, 7 completed 1,000 updates and 3 stopped at numerical guards. None of the completed trajectories met the full-state fixed-point or period-2 recurrence screen; the stopped runs did not reach its final-tail window. Moderate rates memorized the small batch while parameters drifted. This is neither a prevalence estimate nor a test of generalization: validation-corpus outcomes remain unconsumed. Reduced dense-derivative controls and matched-state CPU/GPU comparisons support the matrix-free implementation, without establishing agreement of entire finite-precision trajectories. Loss AR(2) fits can improve one-step prediction by modeling drift, and weak complex fitted roots also appear during ordinary descent. These roots therefore do not establish Floquet multipliers or an NS crossing. The complete evidence and failure labels are in `docs/explore-tiny-lm-receipt.json`; methods and corpus attribution are in `docs/explore-tiny-lm.md`.

Drift-aware prediction pilot (negative calibration). A subsequent development pilot compared causal loss/AR and gradient-history predictors with nonlinear update-plane coherence. The matched-history baseline received all Gram histories needed to reconstruct the coherence features, so these features could only add a nonlinear representation of the same information.

The declared development gate failed: late-event coverage was insufficient, and adding coherence worsened seed-pooled Brier and average precision on the eventful seed. A loss-history baseline was stronger on its eventful trajectory. This sparse challenge establishes no predictive or controller advantage; it does not exclude useful diagnostics on other workloads. Assessment outcomes and controller stages remained unopened. Source-bound results and their independent saved-data audit are documented in `docs/explore-tiny-lm-calibration.md` and `docs/reviews/tiny-lm-calibration-outcome-review.md`.

Broader-data progress forecast (negative). A separate exploratory unit compares nested batches of 8 and 256 text windows using two new initializations and 800 updates per case. Each initialization shares one initial expanded-objective rate scale across its arms. Broader data reduces memorization and lowers loss on a new disjoint development probe, both at equal updates and at equal processed training targets. These are descriptive contrasts on one fixed text pool; the arms are not independently optimized, and equal token counts do not establish equal wall-clock cost.

Predictors fit earlier trajectory segments and tune on subsequent segments, with a horizon embargo before the late evaluation block. The continuous target is training-CE change over the next 16 updates; the probe does not enter fitting or selection. For the expanded arms in seed order, nonlinear update-plane features (B3) give late normalized MSE 0.2999, 0.5026, versus 0.3051, 0.5153 for the complete matched Gram-history baseline (B2) and 0.1027, 0.0949 for the tuning-selected ordinary comparators (loss/gradient history and AR1 respectively). Normalization uses each arm's fit-only target RMS. B3's small improvement over B2 misses the declared improvement threshold and is substantially worse than the ordinary controls; the forecast gate fails for both seeds. An independent saved-data audit reconstructs all candidate fits, selections, forecasts and gates, directly checks corpus/token identities, and replays fixed sampled features. Probe observations are hash-bound and summarized without independent probe reevaluation. No assessment, policy or inference stage follows. This supplies no geometry-specific language-model advantage and does not exclude other useful diagnostics. See `docs/explore-tiny-lm-broader-outcome.md` and `docs/reviews/tiny-lm-broader-outcome-audit.md`.

7 Discussion

Everything in this section is interpretation. The confirmed results in Section 4 stand on their own, and the hypotheses here need their own prespecified tests.

The local part of the thesis now rests on several independent kinds of evidence: the crossing multipliers, the normal-form coefficient and its dynamical checks, agreement of sampled rotation numbers with corresponding crossing angles in two maps, and the same behaviour in a real-data network and in a solvable linear network. The selected scalar case also has a local computer-assisted proof. The picture that fits is the quasi-periodic organization familiar from circle maps and forced oscillators [3, 9, 17]: a closed period-2 orbit can lose stability at a supercritical NS crossing; under the local nondegeneracy assumptions, an attracting invariant circle is born, with rotation approaching the crossing angle. Rational rotations can lock; and the circle may eventually break; this last step remains untested here. The global part, that the periodic windows of the learning-rate plane are resonance tongues of this bifurcation, is supported by the Birkhoff test and by the post-hoc tongue scaling, but its frozen test did not pass. The frozen estimator mixed non-asymptotic and asymptotic levels; a confirmatory version should prespecify an asymptotic-regime estimator, and connecting a tongue to a distant window needs two-parameter fold-curve continuation.

An early spectral diagnostic of learning dynamics is that of van der Maas et al. [36], who

used broadband spectra to recognize chaotic learning; here a spectral peak carries the rotation number, and an autoregressive fit estimates the complex multiplier before the crossing in the tested closed-orbit references.

Open questions. The odd-period windows did not pass the Birkhoff diagnostic; whether they are organized as in periodicity hubs [19] is open. Fixed-start maps mix local criticality with basin accessibility from one shared initialization, as the divergent arc shows, so statements about trainability maps need both. In the real-data network only one of the examined period-2 branches crossed by complex multipliers, and how common such branches are in larger networks is unknown.

What would constitute transfer to language models? The practical hypothesis is that a few unstable or weakly damped modes organize training even when parameter dimension is large. Matrix-free Jacobian–vector products through the complete optimizer update can estimate those modes without constructing a dense Jacobian. This hypothesis needs tests of numerical residuals, mode coverage, checkpoint-to-checkpoint drift and predictive accuracy. Along nonperiodic or scheduled training, finite-horizon tangent growth is the appropriate initial quantity; Floquet language is reserved for a sufficiently closed periodic reference orbit. The fixed-batch transformer pilot establishes an architectural test path, but has not yielded a periodic reference orbit. Neither the drift-aware event calibration nor the separate broader-data continuous-progress pilot has shown incremental predictive utility. Separate minibatch, optimizer-state, schedule and precision perturbations remain necessary. Relevant endpoints are prospective transition predictions and improved interventions at comparable compute cost. Similar-looking loss oscillations would not establish either endpoint. A linear loss-based multiplier estimate alone also does not identify nonlinear criticality or basin accessibility.

Inference is a separate question. The bifurcations studied here concern optimizer updates in parameter space. Freezing weights stops that map; sensitivity of fixed-weight predictions and feedback during generation require separate measurements. Parameter symmetries can even leave predictions unchanged along weight motion. A future inference test could ask whether training diagnostics predict sensitivity to arithmetic or weight perturbations beyond loss and token-score margins. For a unique greedy choice with logit gap m, a uniform logit-error bound $\epsilon < m/2$ preserves that choice at the given prefix. Empirical precision comparisons are not rigorous error bounds, and divergent generated text alone does not establish chaos. No inference experiment or benefit is reported here.

The tests that would settle the local claims. The criticality prediction is a candidate for confirmation on fresh losses: the sign of d at each crossing predicts a circle of normal-form amplitude beyond a supercritical edge, and a local unstable-circle scale before a subcritical crossing. The distant state, loss spike, recovery and hysteresis also depend on global basins, ramp speed and noise and must be tested separately. An existence-and-replication test would apply this, together with the onset and rotation criteria, to two prespecified independent losses across two architectures, one of them unused so far, with no substitution. The ancestry test additionally needs a prespecified asymptotic-regime estimator, reliable continuation of both fold edges, and rules for unresolved or disconnected branches before protected holdouts are opened.

8 Limitations

The map study uses networks of 13 and 72 parameters on synthetic data; the real-data result is one branch of one 529-parameter network on one subset and task; the linear network is a solvable model. These map and bifurcation experiments use deterministic full-batch float64 training. The transformer development pilot uses float32 with matched-state float64 checks and has no accepted periodic orbit. Screening labels are finite-time; chaotic labels rest on single-tangent exponents and reruns. The orbit census uses floating-point validation; the three selected exact-real attracting orbits and the scalar local NS in Section 5 have rigorous enclosures. The first Lyapunov coefficient is a local, third-order quantity; it predicts the birth of a small circle, not how far it survives. Rotation numbers and the Birkhoff test use two-dimensional projections of curves in many dimensions. Merged crossing loci combine independently initialized output slices and do not establish branch identity between them or generalized-NS points. Most mechanism analyses were chosen after seeing related outcomes. The consequence pilot measures held-out loss and perturbation response in small models, but establishes no robust training advantage. Transfer to stochastic training, modern language models and useful interventions remains open.

9 Related work

Edge of stability. Cohen et al. [6] describe the edge of stability; Damian et al. [10] explain self-stabilization of sharpness at the edge, whereas we follow what happens after the period-2 oscillation itself loses stability; Agarwala et al. [1] show progressive sharpening to the edge in second-order regression models, the approach to the regime we study beyond it; Lewkowycz et al. [24] describe a catapult phase at large rates, in which the loss first grows and then settles in flatter regions, while our maps resolve the long-run dynamics of rates where training does not settle. Hofmann [15] derives the rotational edge whose far side we study. Chen and Bruna [4] analyze two-step GD, proving period-2 convergence in specified scalar and single-neuron settings and testing related conditions in larger models. Song and Yun [33] study alignment with a reduced bifurcation diagram, with proofs for specified single-example models and neural-network experiments. Our orbit tests concern closure and stability of the full parameter state. Further analyses include Kalra et al. [20]. Chen et al. [5] reduce specified quadratic regression settings to a cubic map; their Li–Yorke chaotic regime can also contain attracting periodic orbits. Cohen et al. [8] model time-averaged deterministic GD and RMSProp trajectories with central flows, with explicit limitations on rigor, stochastic training, momentum and computational feasibility. Our full-state orbit and boundary tests complement this averaged description.

Adaptive optimizers and larger models. Cohen et al. [7] study full-batch and large-batch adaptive methods at a preconditioned stability threshold. Regis and Chewi [30] model Adam in the joint parameter–first-moment state, with the second moment as an auxiliary variable, and compare rod-flow approximations across architectures and optimizers. Fong and Yang [11] connect second-moment feedback to active curvature and report gradient reversal, including in GPT-2 medium. The latter uses a fixed text batch and describes approximate alternating motion around a drifting center; that description is distinct from full-state periodic-orbit closure. Its minibatch experiments also show that noise and momentum can weaken the reversal. Cohen et al. already examine transformer warmup, while the rod-flow study explicitly limits its model to full-batch steady-state behavior. These studies establish relevant prior work on oscillatory training in larger models. Our proposed extension must identify an additional mechanism or predictive consequence;

detecting oscillation in a transformer alone would not supply that contribution.

Chaos and fractals in training. Kong and Tao [21] analyze deterministic GD on multiscale objectives. Jiménez-González et al. [18] relate finite-time perturbation expansion to epochs needed to reach task-specific accuracy thresholds in full-batch MLP/CNN experiments. This is relevant prior utility evidence; our proposed geometry-specific benefit needs an additional comparison. Sohl-Dickstein [32] study fractal trainability boundaries; Torkamandi [35] map loss-based convergence boundaries in a small character transformer trained with Adam; Liu [26] study learning-rate boundaries for cosine-perturbed quadratic objectives. Liang and Montúfar [25] analyze initialization-space convergence regions in regularized factorization, with orthogonality restrictions for their matrix extension. Ly and Gong [27] report empirical riddling-like and functional sensitivity across training runs, including BERT fine-tuning; this concerns outcomes after retraining from nearby initializations, not chaos during frozen inference. Learning dynamics in games show chaos [31] and phase locking [28], for adaptive rules rather than descent on one loss.

Dynamical systems. Shrimp-shaped windows are described by Gallas [12, 13]; circle maps, Arnold tongues, devil's staircases and the breakdown of invariant circles by Arnold [2], Aronson et al. [3], Curry and Yorke [9], Jensen et al. [16, 17], Ostlund et al. [29]; numerical bifurcation analysis of maps, by Kuznetsov and Meijer [22], with the local NS theorem and coefficient stated by Kuznetsov and Sacker [23]. Our contribution is the numerical characterization and testing of these mechanisms in the specified training maps. A literature search cannot establish their absence from all earlier optimizer studies.

10 Reproducibility

Code, plans and compact receipts live in the project repository; bulky raw maps live in a separate data store referenced by SHA-256. Every figure is generated by a script in `paper/gd-route/figures/` that reads saved evidence and writes a receipt with input hashes, the generator's hash, output hashes, alt text and the values it displays. Quoted empirical numbers and the design and consequence tables are generated by `paper/gd-route/scripts/build_numbers.py`; `data/numbers.json` records each number's source file, hash and computation, and `provenance.json` indexes all receipts. `paper/gd-route/scripts/verify.py` regenerates every figure and quoted number, replays the scalar-mechanism, warmup and initial transformer feasibility receipts, orbit certificates and scalar NS gates, and fails on any difference. Later S1R, CIFAR and LM pilots have separate source-bound saved-data audits cited with their results. The verification status of every reference is kept in `data/references-ledger.json`. The map study corresponds to the repository's EXP-028 plan and audit; its plan, receipts and audit are linked from `docs/EXP-028-results.md`. The exploratory streams are summarized in `docs/explore-criticality-2026-10-07.md`, `docs/explore-ancestry-2026-10-07.md`, `docs/explore-realdata-2026-10-07.md`, `docs/explore-minimal-model-2026-10-07.md` and `docs/explore-consequence-2026-10-07.md`.

Placeholder. Author contributions, acknowledgments, license choice and public code and data links are human decisions to be made before release.

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